AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 13 · STUDENT LESSON

Learning from Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Compare candidates. Preserve the tie.

Fit tower and stick rules, inspect unresolved predictions and record how an added label changes selection.

Keep one habit: Trace predictions → count each candidate's errors → distinguish a choice from new evidence.

MAIN LESSON · SECTION 1

What did the fitting examples actually decide?

> Trace each candidate, keep separate error totals and record unresolved ties.

Start with equal fitting scores

Two rules can agree on every fitting example and still disagree on a new input. A declared tie rule chooses reproducibly; it does not supply the missing label.

Retrieve the equality check

A rule predicts tall when height is at least 5 cm. What does it predict at 4, 5 and 6 cm? Why is 5 a useful boundary test?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can fit a threshold by tracing **prediction**s and counting disagreements for every candidate.
- I can apply an explicit **tie rule** while recording what the **fitting** examples leave unresolved.
- I can use a selected rule on a **new input** and explain why fit alone does not establish future correctness.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Training examples influence how a model handles later **input**s.
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LEARNING TARGET **Training** cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can compare **candidate model**s, resolve a tie explicitly and separate fit from **prediction**.

MAIN LESSON · SECTION 2

State the fitting procedure before comparing results

> The tower labels belong to this fictional task and use height as the only feature.

The supplied towers and labels

Four imaginary paper towers have heights 2, 4, 6 and 8 centimetres. Their supplied labels are short, short, tall and tall. These labels belong to this classroom task; they are not a universal definition of tall.

We will fit a rule using height as its only **input**.

The candidates and declared tie rule

Every candidate predicts tall when height is at least its threshold and short otherwise. Compare thresholds 3, 5, 6 and 7 centimetres. The **fitting** procedure chooses the fewest disagreements with the four labels.

If candidates tie, choose the smaller threshold. Write these choices before comparing results.

Count disagreements separately for every candidate

For each candidate, start its count at zero. For each example, predict with that candidate, compare with the supplied label, and add one only for a disagreement. After all four examples, record the count.

Then move to the next candidate with a fresh count. This uses the conditions and repetition you have already traced.

Prepare the comparison

- Before calculating, write the candidate thresholds, comparison phrase and tie policy. For each candidate, where does its error count begin, and when should you reset that count?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A learner shown mostly striped triangles may wrongly treat stripes as the important clue for the triangle label.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Check each error against the example that caused it

> A zero count is an observed fitting result with a clear denominator.

Compare the four candidates

Disagreements on the four fitting examples

CANDIDATE	DISAGREEMENTS
Threshold 3	1 of 4
Threshold 5	0 of 4
Threshold 6	0 of 4
Threshold 7	1 of 4

All candidates use the same four examples and the same zero to four scale. Thresholds 5 and 6 have zero-length bars. They tie; our stated smaller-threshold rule selects 5.

Locate the mismatched tower rows

Threshold 3 incorrectly calls the height 4 tower tall. Threshold 7 incorrectly calls the height 6 tower short. Thresholds 5 and 6 agree with all four supplied labels.

Check each disagreement against the individual example instead of trusting a total without its trace.

Read the count and its limit

The bar chart compares **disagreement count**s, so a shorter bar is better on this **fitting set**. Zero is a measured count here, not missing data. This chart says nothing yet about the error count on new towers.

Explain a chart entry

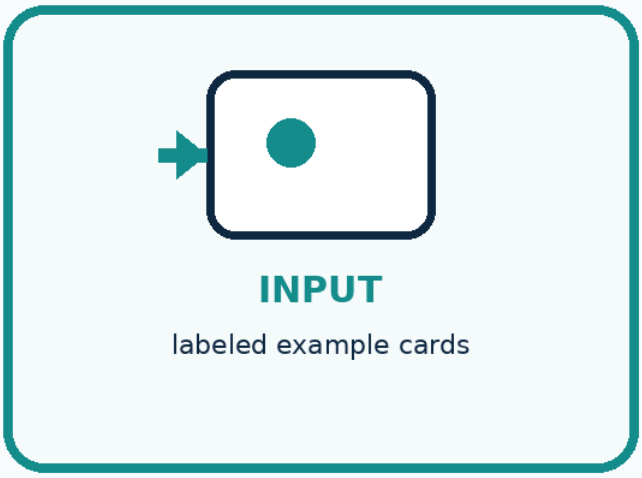
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

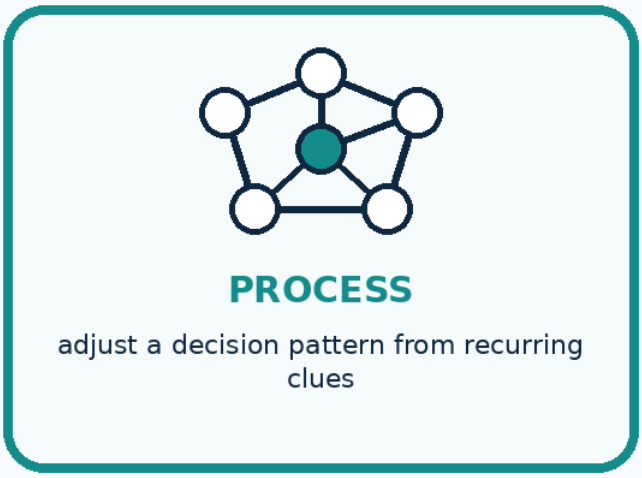
▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT labeled example cards

PROCESS adjust a decision **pattern** from recurring clues

OUTPUT prediction for a new card

CORE IDEA **Training example**s influence how a model handles later **input**s.

UNDER THE HOOD Supervised learning uses labeled examples to fit a mapping from represented **input**s to **output**s.

The model does not store a human rule automatically; its fitted behavior reflects correlations, coverage, labels, and optimization choices.

Draw a fresh example with all three stages:

A closer explanation

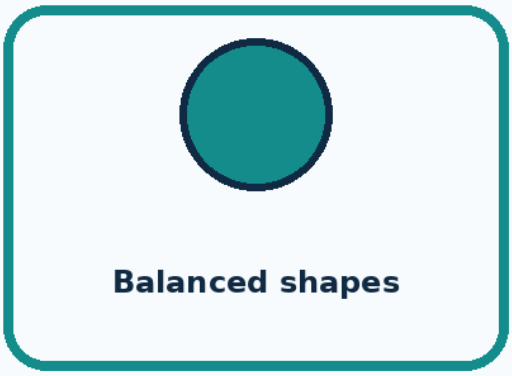
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare fitting, prediction and evidence gaps

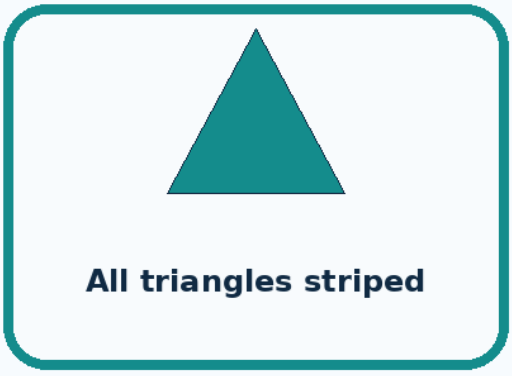
> Use the supplied examples and ask which observations would distinguish a proposed rule.

Pictures paired with their evidence



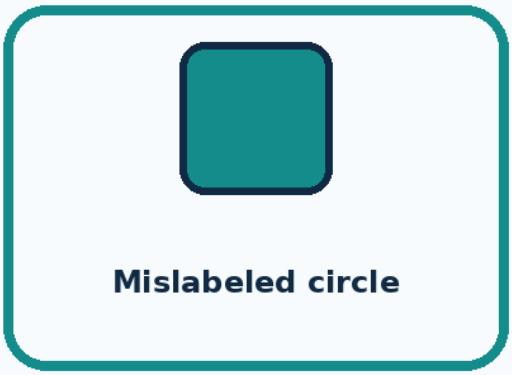
The case to investigate

CASE 1 — Balanced shapes Clue: colors vary within each shape. Meaning: shape clue becomes clearer.



The case to investigate

CASE 2 — All triangles striped Clue: stripe and triangle always co-occur. Meaning: confounded clue.



The case to investigate

CASE 3 — Mislabeled circle Clue: target is incorrect. Meaning: label noise.



The case to investigate

CASE 4 — New dotted triangle Clue: unseen combination. Meaning: tests learned behavior.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

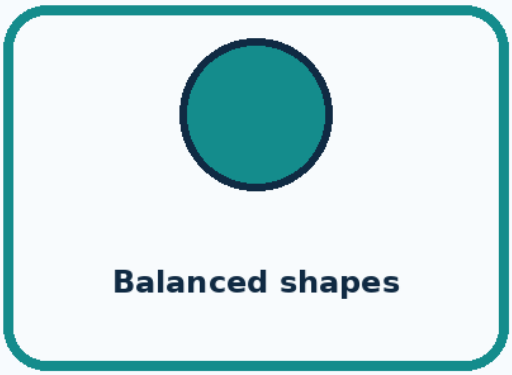
BOUNDARY Learning from examples is not the same as memorizing every example or understanding the concept like a person.

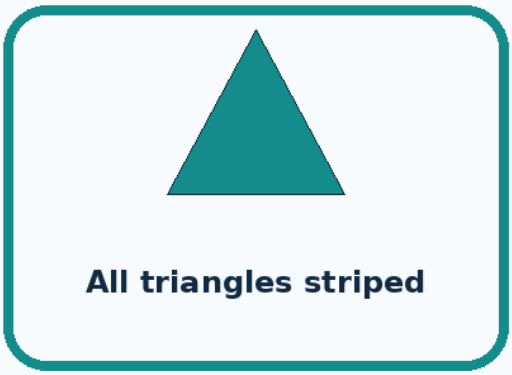
▼ **Earlier wording, alternatives and picture transcripts**

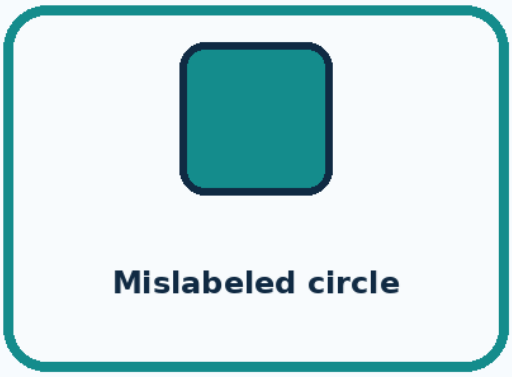
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Balanced shapes Clue: colors vary within each shape. Meaning: shape clue becomes clearer.

CASE 2 — All triangles striped Clue: stripe and triangle always co-occur. Meaning: confounded clue.

CASE 3 — Mislabeled circle Clue: target is incorrect. Meaning: label noise.

CASE 4 — New dotted triangle Clue: unseen combination. Meaning: tests learned behavior.

Change one thing in the pictured case

Change: Reverse the tie policy to prefer the larger boundary.

Prediction: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

MAIN LESSON · SECTION 6

A tie policy is a choice, not a new observation

> Use the stick task's boundaries 5 and 7 and its stated smaller-boundary tie rule.

The fitting sticks, candidates and policy

A fictional classifier learns whether measured sticks are short or long.

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

Compare the tied candidates

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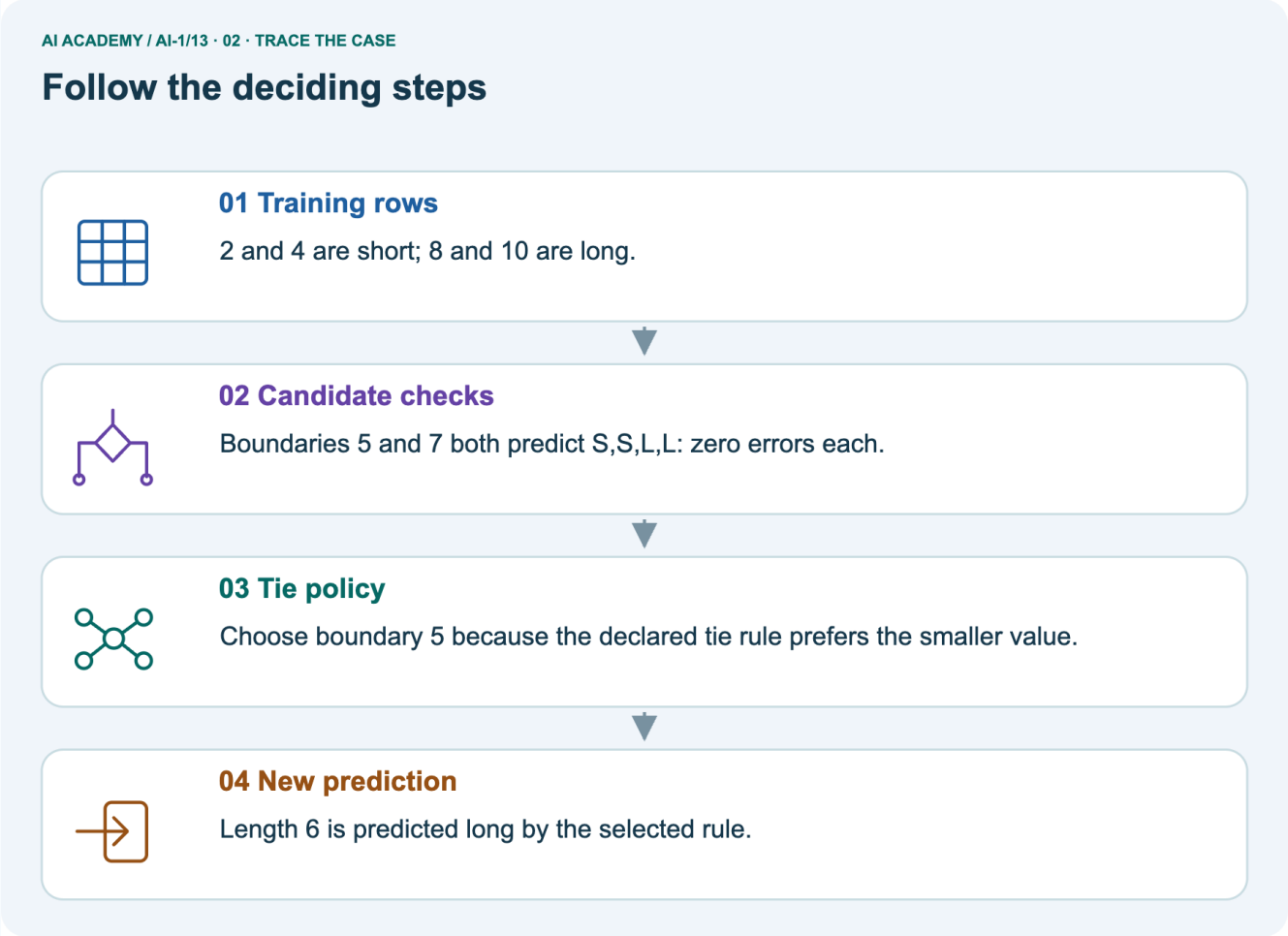
Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace selection and label-free prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the unresolved interval

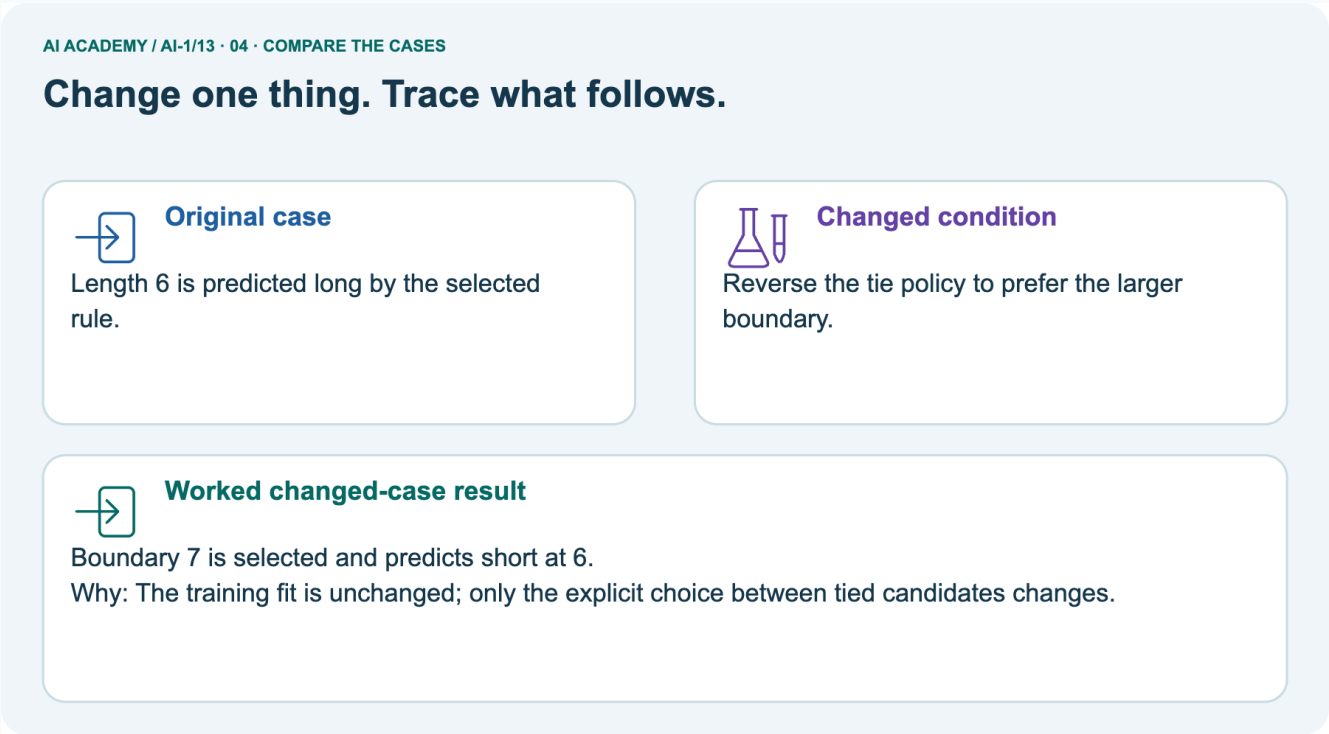
There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label.

Predict with the opposite tie policy

- Before checking, keep the fitting sticks and candidate boundaries unchanged but prefer the larger boundary in a tie. Which model is selected, what does it predict at length 6, and which evidence has not changed?

▼ Check the changed selection



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the zero-error claim

Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

Choose a length where the models disagree

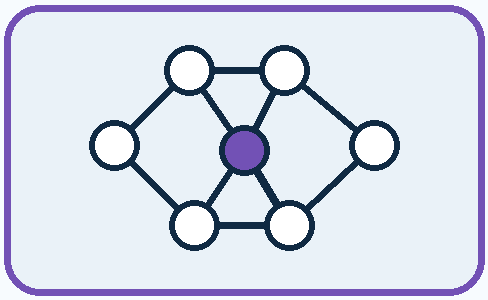
- Read the labelled picture: Which new measured length would reveal disagreement between the candidates?

▼ **Earlier wording, alternatives and picture transcripts**

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A learner shown mostly striped triangles may wrongly treat stripes as the important clue for the triangle label.

STEP 1 — OBSERVE labeled example cards

STEP 2 — TRACE adjust a decision **pattern** from recurring clues

STEP 3 — CONCLUDE **prediction** for a new card

MODEL REASONING Early examples may support several rules. New examples eliminate some possibilities and may change the decision boundary. Record each revision as **evidence** that **training** data shapes later behavior.

DO NOT OVERCLAIM Learning from examples is not the same as memorizing every example or understanding the concept like a person.

What new **evidence** could change this answer?

A closer explanation

Height and supplied label	Prediction under threshold 5	Prediction under threshold 6
2 cm short	Short	Short
4 cm short	Short	Short
6 cm tall	Tall	Tall
8 cm tall	Tall	Tall
5 cm label not yet supplied	Tall	Short

The first four rows are **fitting** examples. The final row asks what the two tied candidates predict at a height missing from that set. Our **tie rule** selects 5, so the selected model predicts tall at 5.

The **fitting** examples alone do not tell us whether that new **prediction** matches the task label.

A **tie rule** makes the selection reproducible. It does not prove that the chosen threshold is the true boundary. Record that 6 tied with 5 and that they disagree at height 5.

Hiding the tie would hide a real limitation of the available **evidence**.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning from Examples

The facts we will use

AI ACADEMY / AI-1/13

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Learning from Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training rows

2 and 4 are short; 8 and 10 are long.

2. Candidate checks

Boundaries 5 and 7 both predict S,S,L,L: zero errors each.

3. Tie policy

Choose boundary 5 because the declared tie rule prefers the smaller value.

4. New prediction

Length 6 is predicted long by the selected rule.

CHANGE ONE THING

Reverse the tie policy to prefer the larger boundary.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training rows	2 and 4 are short; 8 and 10 are long.
2. Candidate checks	Boundaries 5 and 7 both predict S,S,L,L: zero errors each.
3. Tie policy	Choose boundary 5 because the declared tie rule prefers the smaller value.
4. New prediction	Length 6 is predicted long by the selected rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Reverse the tie policy to prefer the larger boundary. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Best among these candidates has a limited meaning

> Features, candidate families and label policies remain human choices.

Keep the comparison scope explicit

The best candidate means best among the candidates compared, using this counting rule and these recorded labels. A different candidate family, feature or label policy might give a different result.

People chose those ingredients; **fitting** does not remove those choices.

Inspect conflicting labels for identical inputs

If two records with identical **input** height have different labels, any deterministic rule using only that height gives them the same **prediction**. It must disagree with at least one label.

Investigate the records and task definition instead of silently changing a label to make the model look perfect.

Repair an unsupported future guarantee

- CLAIM A The model discovers the one true rule hidden in nature.
- CLAIM B One example is enough to learn a reliable category.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **The selected threshold has zero disagreements on four fitting examples. A learner claims it will label every future tower correctly. Write a more accurate claim and name evidence that is still needed**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Balanced shapes	colors vary within each shape	shape clue becomes clearer
All triangles striped	stripe and triangle always co-occur	confounded clue
Mislabeled circle	target is incorrect	label noise
New dotted triangle	unseen combination	tests learned behavior

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: add a label and recompute the fit

> Retain the original examples and distinguish fitting evidence from a final check.

The new labelled tower and your comparison

Suppose the task designer supplies an additional **fitting** example: height 5 centimetres, label short. Compare thresholds 5 and 6 again using all five **fitting** examples.

Do not discard the original four when adding the new example.

> **List the prediction at the added height 5 under each candidate and update both total disagreement counts. Select between 5 and 6 under the fitting procedure. Explain why the added example is now fitting evidence rather than an independent final test**

Explain why a selection label becomes fitting data

A label that helps choose the threshold participates in **fitting**. This is allowed, but it changes what the example can establish. We must use separate **evidence** to evaluate the selected model beyond the examples that shaped it.

Week 15 will develop that separation in detail.

Use W2 to inspect the original tied candidates

Use workbook W2 for the connected example "**Fitting** a **pattern** from examples". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge adding error counts across candidates

- Use the worked case for **Fitting** a **pattern** from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Choose a distinguishing stick length

Which new measured length would reveal disagreement between the candidates?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Training example**s influence how a model handles later **input**s.

Use a fresh example to explain **training example**.

Trace a **new input** → **process** → **output** example.

Find and repair the error in: One example is enough to learn a reliable category.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Explain a tied fit

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate how a new labelled shape changes a model

> Run a declared candidate comparison and keep the version change visible.

The fitting cards and complete procedure

Use fictional shape cards whose feature is corner count. The fitting cards are 3 corners labelled FEW and 5 corners labelled MANY. Candidate 4 predicts MANY at 4 or more corners; candidate 5 predicts MANY at 5 or more. Both otherwise predict FEW. Choose the lower error count; a tie prefers the smaller candidate. These labels are task conventions.

Predict, fit and record the new prediction

1. Predict every fitting row, count each candidate's errors and select a model. Before revealing another label, predict the selected model's label for a four-corner card. Record your first prediction. Then open the new-label card below, add it to the fitting record and recompute both candidates. Explain whether that newly used label is still an independent test.

▼ **Open the new fitting label after predicting**

New fitting label for this invented task: the four-corner card is FEW. Add this labelled example without erasing either earlier example. Preserve the old model, first prediction, new counts and new selected model as separate records.

Compare the model versions and limits

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your **prediction**.

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

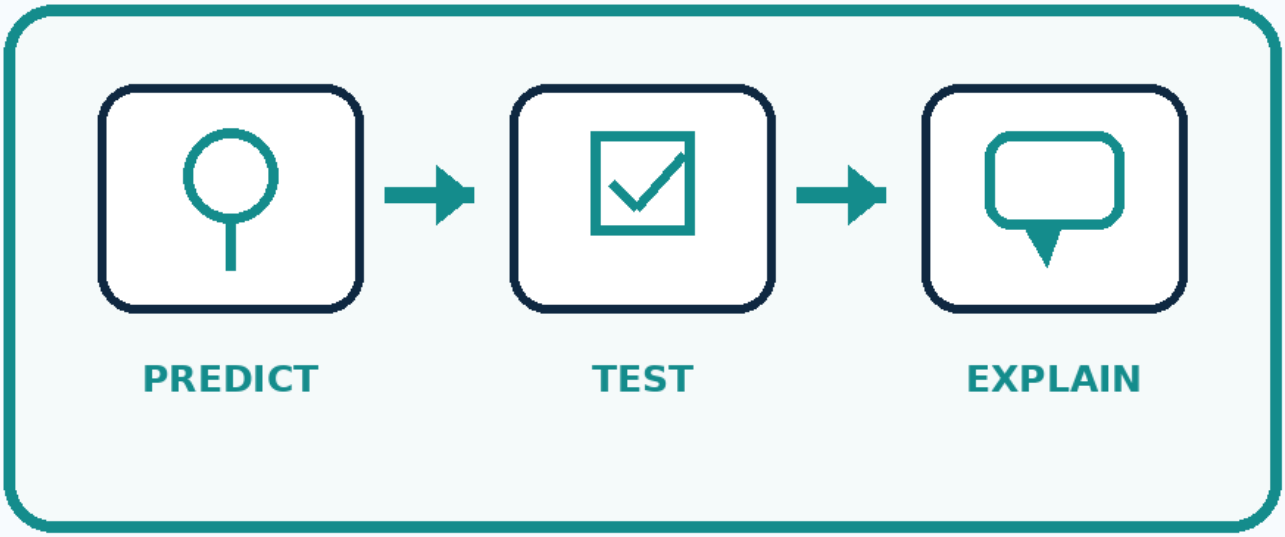
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Human Shape Model

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Act as a model trained on shape cards; record rule changes.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit a new distance task

> Use its supplied near/far labels and candidate distances before seeing a label for the new input.

The new fitting task

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Fit a new distance task

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing **evidence** could help? How would you check improvement?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create the first small labeled example set for the sorter and explain which visual differences it covers.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the full fitting decision with your sorter model

> Keep candidate definitions, separate errors and tie policy beside the selected rule.

This week’s project decision

THIS WEEK'S CHECKPOINT Create the first small labeled example set for the sorter and explain which visual differences it covers.

Save candidate definitions, **training** errors and tie policy with any sorter model version.

Plan examples, checks and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep learning distinguishable from a human understanding claim

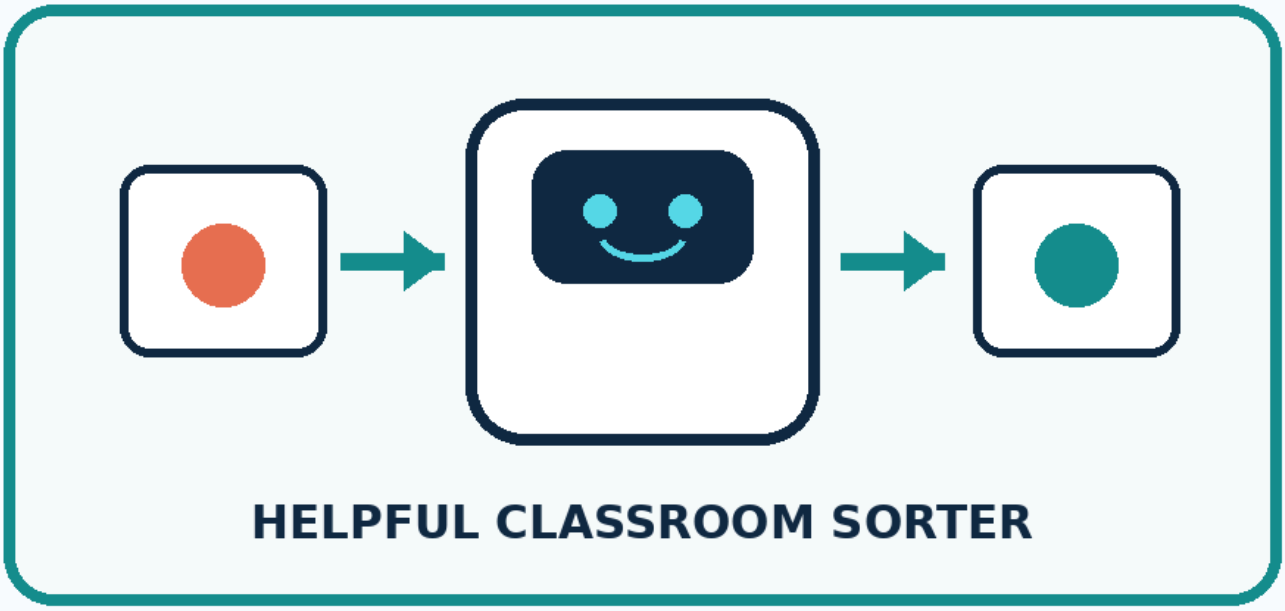
RESPONSIBLE DESIGN Learning from examples is not the same as memorizing every example or understanding the concept like a person.

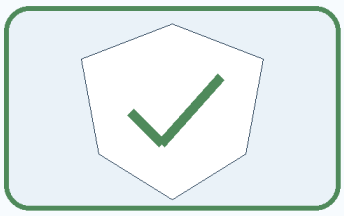
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain selection without overstating the fit

> Close your notes and distinguish the chosen rule, its prediction and evidence of correctness.

Your exit explanation

- Close your notes. Explain this goal in your own words: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
training example	input paired with a target label	
pattern	recurring relationship across examples	
training	fitting behavior from examples	
new input	case not used in fitting	

Terms for the connected explanation

Word	Meaning
Candidate model	One possible rule being compared during fitting
Fitting set	The examples used to choose or adjust a model
Disagreement count	The number of predictions that differ from their comparison labels

Word	Meaning
Tie rule	An explicit method for choosing between equally scoring candidates
Generalisation	How well a fitted model works on new cases beyond its fitting examples

Repair a claim using a word

- CLAIM TO REPAIR The model discovers the one true rule hidden in nature.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

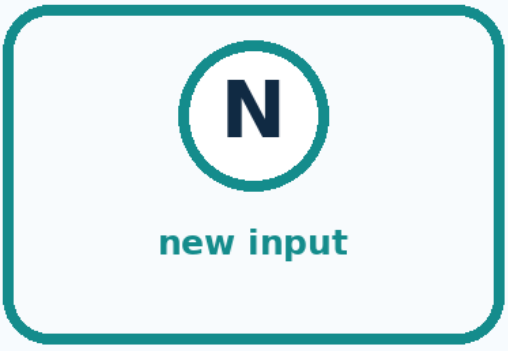
Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

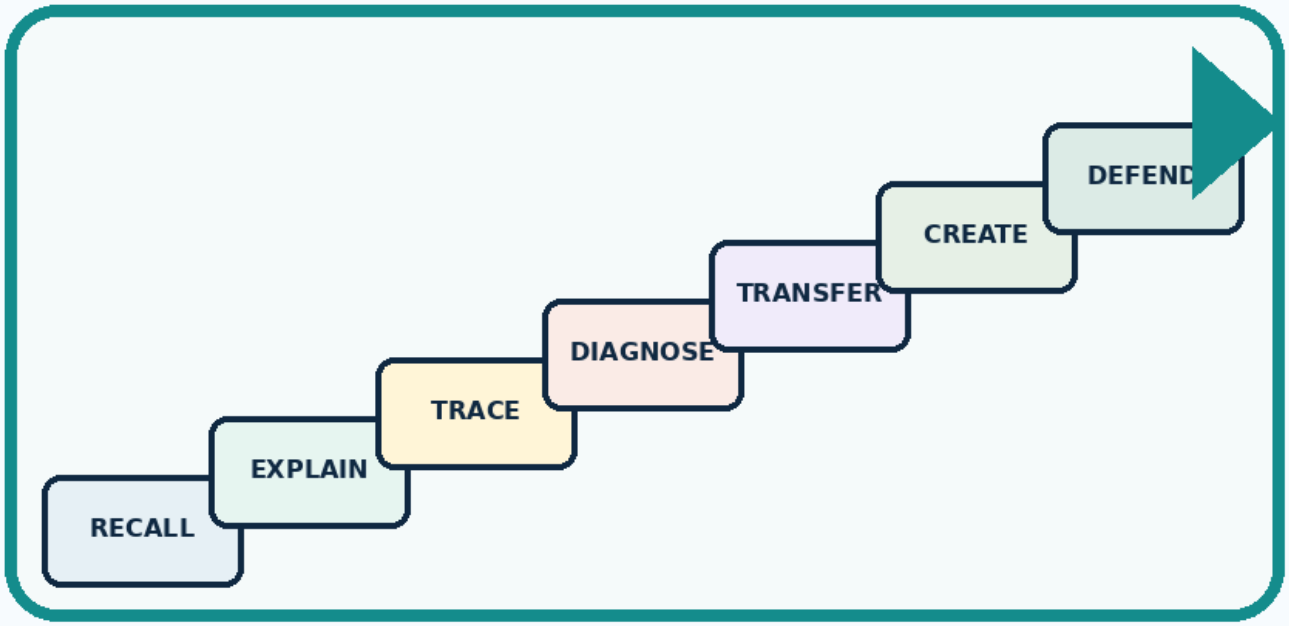
- RECALL Define **training example**.
- EXPLAIN **Training example**s influence how a model handles later **input**s.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The model discovers the one true rule hidden in nature.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Fit a new distance task is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Fit a new distance task is the selected independent case.



EXPLANATION CHECK List predictions and separate disagreement counts, apply the declared selection and tie rules, and state what the fitting set leaves untested.

Independent **evidence**:

SELF-CHECK Early examples may support several rules. New examples eliminate some possibilities and may change the decision boundary. Record each revision as **evidence** that **training** data shapes later behavior.

NEXT BRIDGE Carry today's **evidence** habit into: Build a Human Classifier.

▼ **Extra practice: Examples suggest a pattern to test**

REFERENCE · SECTION 13

Extra practice: Examples suggest a pattern to test

> Try the question before opening its worked explanation.

Read the supplied case and question

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the **evidence** limit.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Predict T from the demonstrated edge **pattern**. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

Evidence or stage	Value or result
Smooth leaves	Training label S
Toothed leaves	Training label T
New toothed leaf	Predict T
Fresh labelled cases	Check the prediction rule

▼ Earlier wording, alternatives and picture transcripts

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Examples suggest a pattern to test

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the **evidence** limit.

Predict T from the demonstrated edge **pattern**. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Fit a threshold by tracing predictions and counting disagreements for every candidate	Explain the deciding step, show a trace or test, and state one limit.
Apply an explicit tie rule while recording what the fitting examples leave unresolved	Explain the deciding step, show a trace or test, and state one limit.
Use a selected rule on a new input and explain why fit alone does not establish future correctness	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional classifier learns whether measured sticks are short or long.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classifier learns whether measured sticks are short or long.
- Which labelled step matches this detail? Length 6 is predicted long by the selected rule.
- 1. **Training** rows
- 2. New **prediction**
- 3. Tie policy
- 4. Candidate checks
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: New **prediction** Length 6 is predicted long by the selected rule. There is no **training** stick between 4 and 8 to distinguish these candidates.

The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true. Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label.

Changed-case question

- Changed-case practice: Reverse the tie policy to prefer the larger boundary. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Boundary 7 is selected and predicts short at 6.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Boundary 7 is selected and predicts short at 6. The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Repair and save your work

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

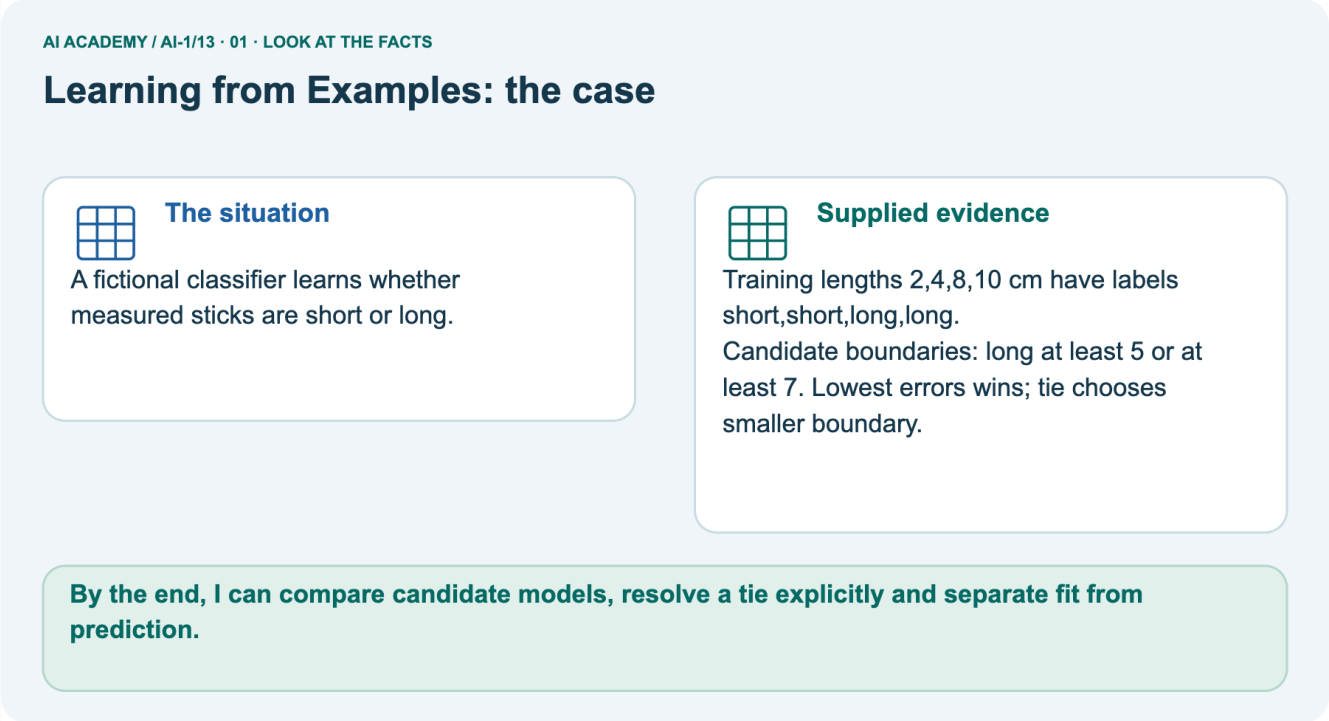
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Learning from Examples: the case



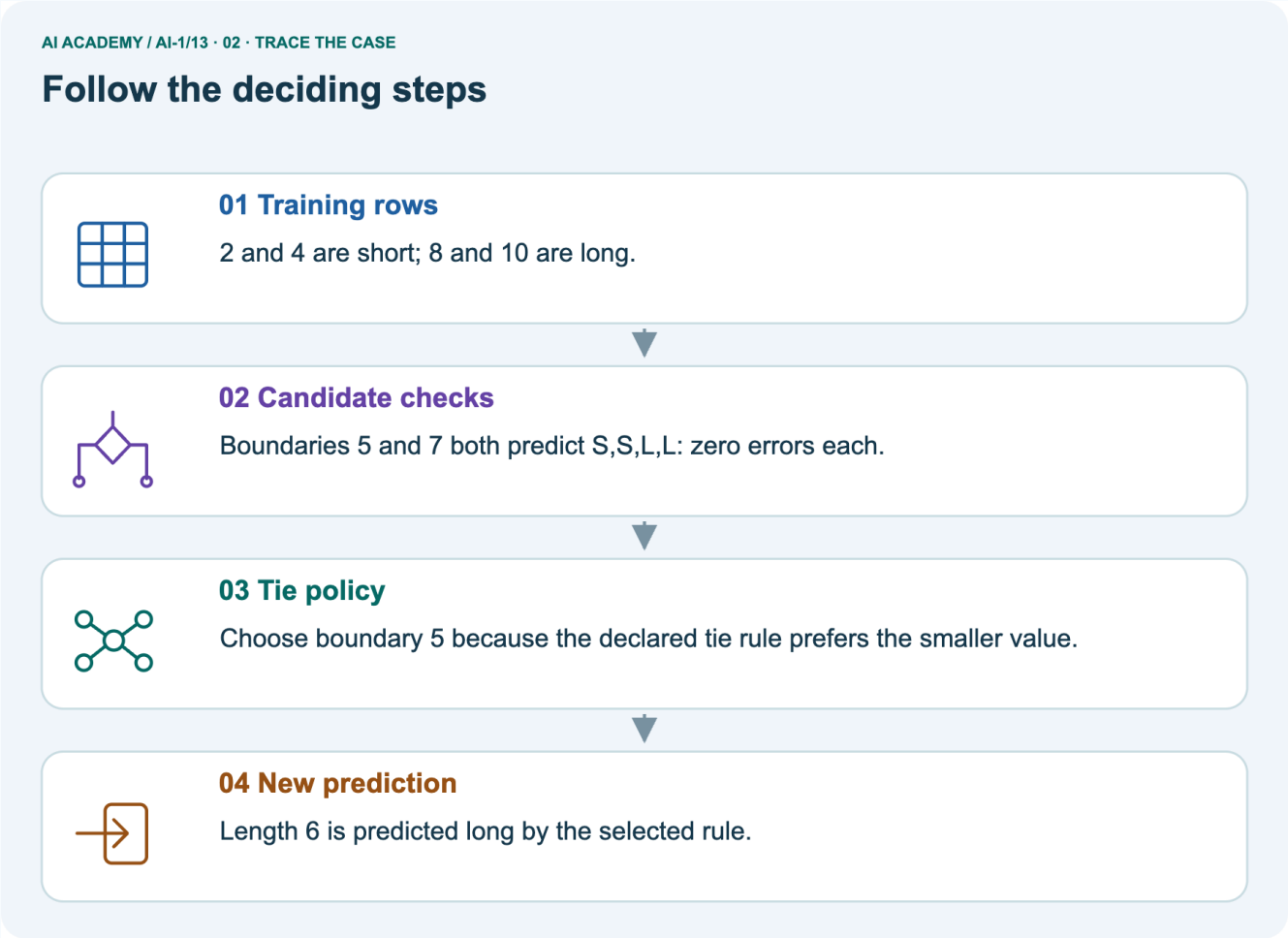
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classifier learns whether measured sticks are short or long. Supplied facts: **Training** lengths 2,4,8,10 cm have labels short,short,long,long. Candidate boundaries: long at least 5 or at least 7.

Lowest errors wins; tie chooses smaller boundary.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

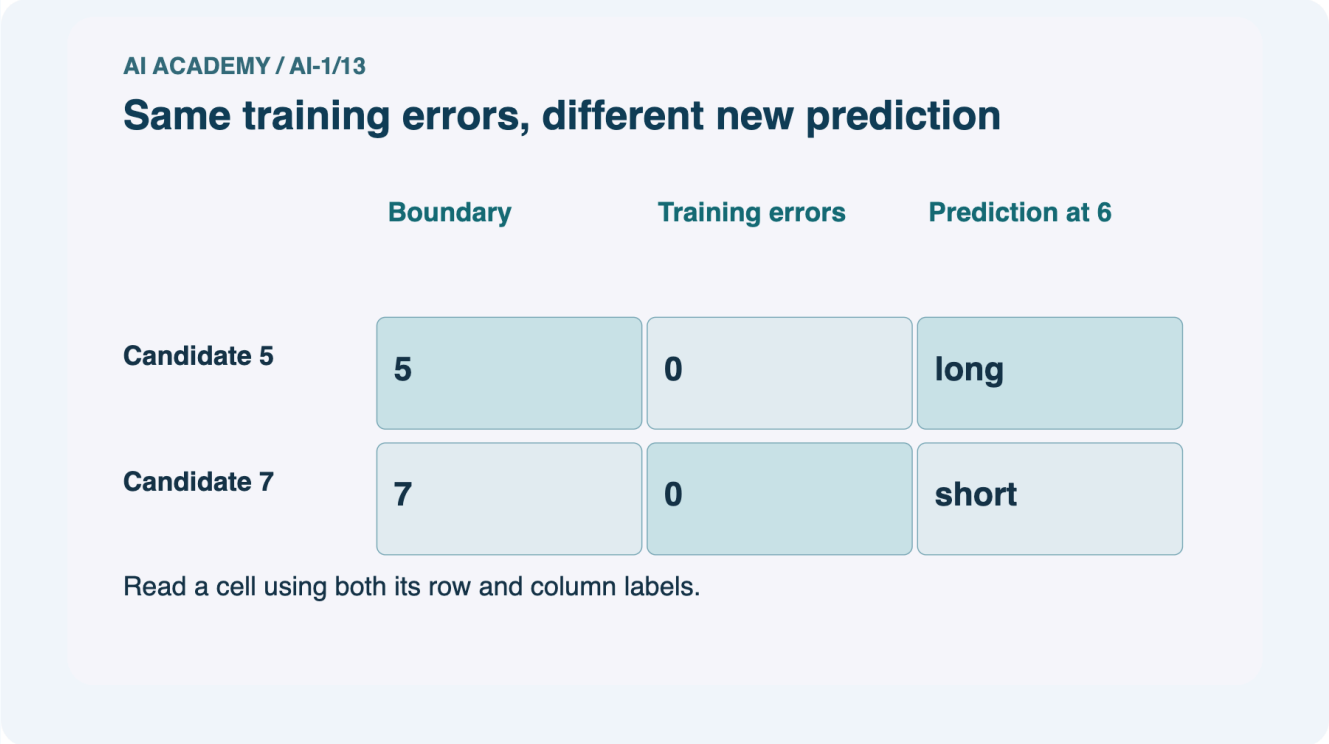


Follow the labelled picture with the matching explanation. Open the vector diagram

Training rows: 2 and 4 are short; 8 and 10 are long. → Candidate checks: Boundaries 5 and 7 both predict S,S,L,L: zero errors each. → Tie policy: Choose boundary 5 because the declared **tie rule** prefers the smaller value. → New **prediction**: Length 6 is predicted long by the selected rule.

Point to each numbered stage and explain which recorded **input** it uses.

03. Same training errors, different new prediction

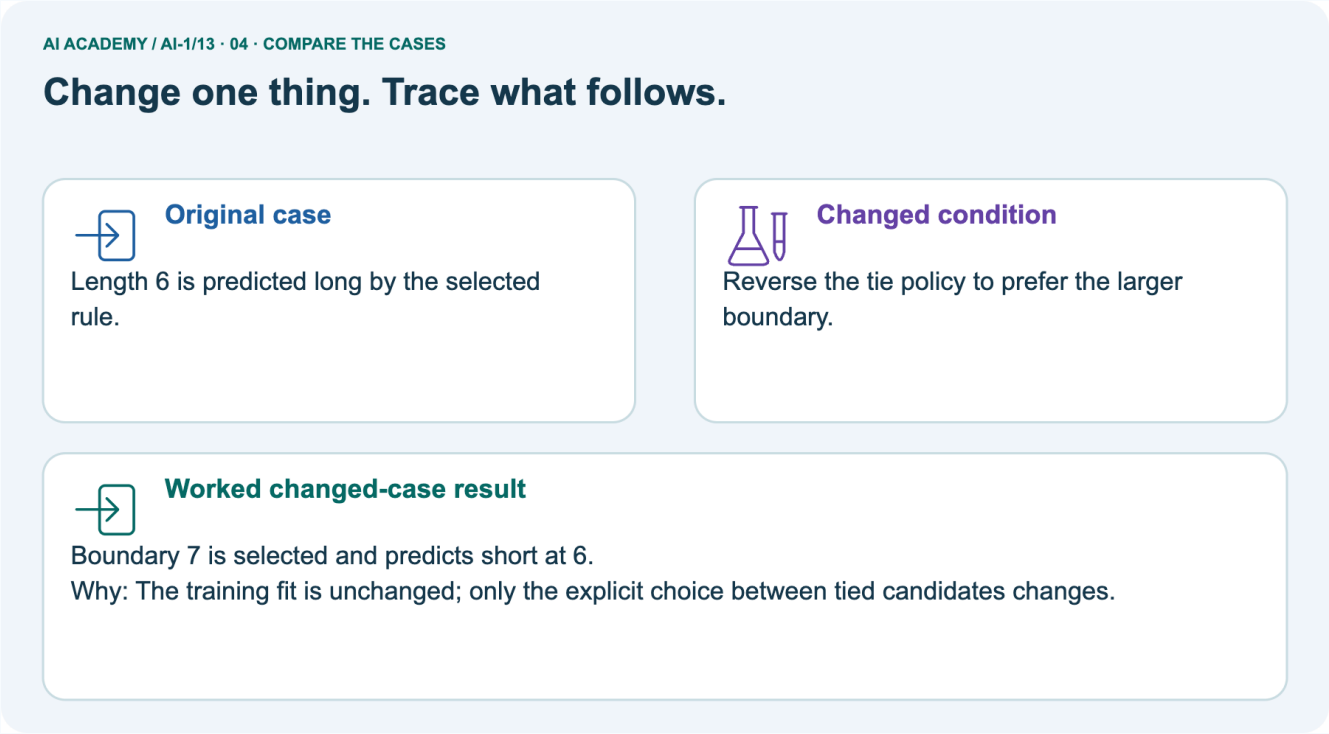


Follow the labelled picture with the matching explanation. Open the vector diagram

Same **training** errors, different new **prediction**. Candidate 5: Boundary = 5, **Training** errors = 0, **Prediction** at 6 = long. Candidate 7: Boundary = 7, **Training** errors = 0, **Prediction** at 6 = short.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



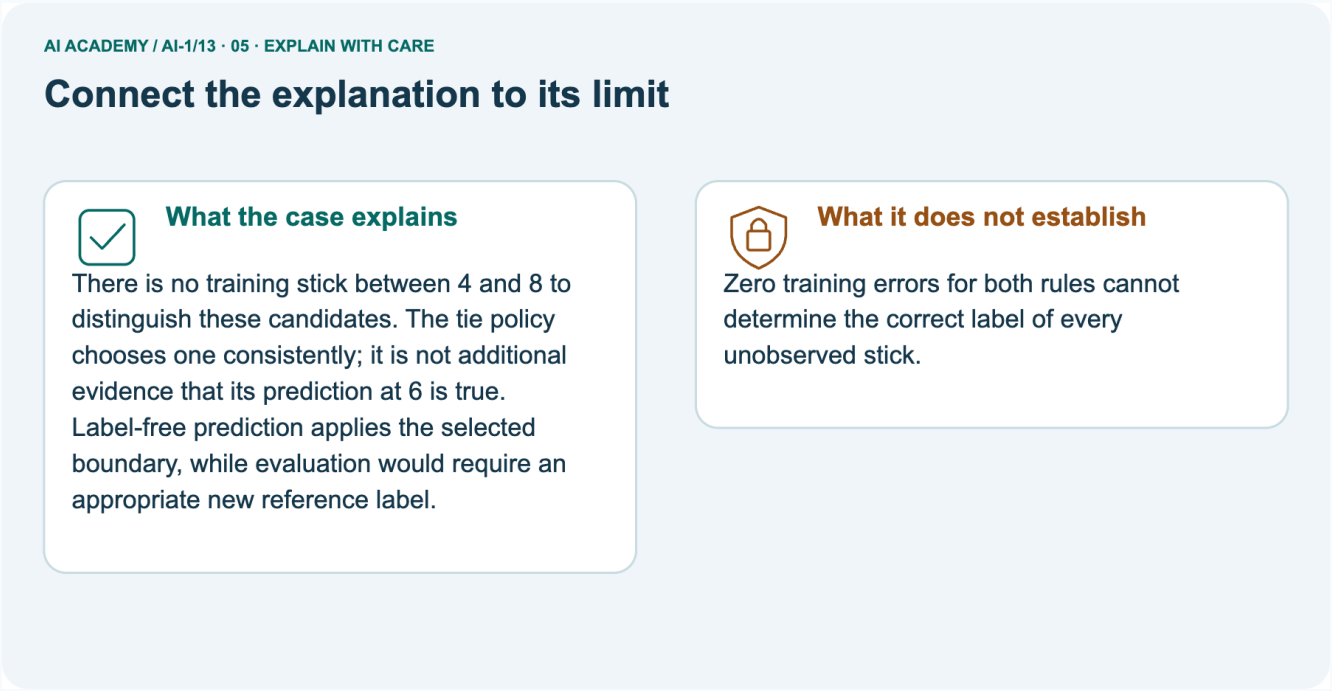
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Length 6 is predicted long by the selected rule. Changed condition: Reverse the tie policy to prefer the larger boundary. Expected result: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



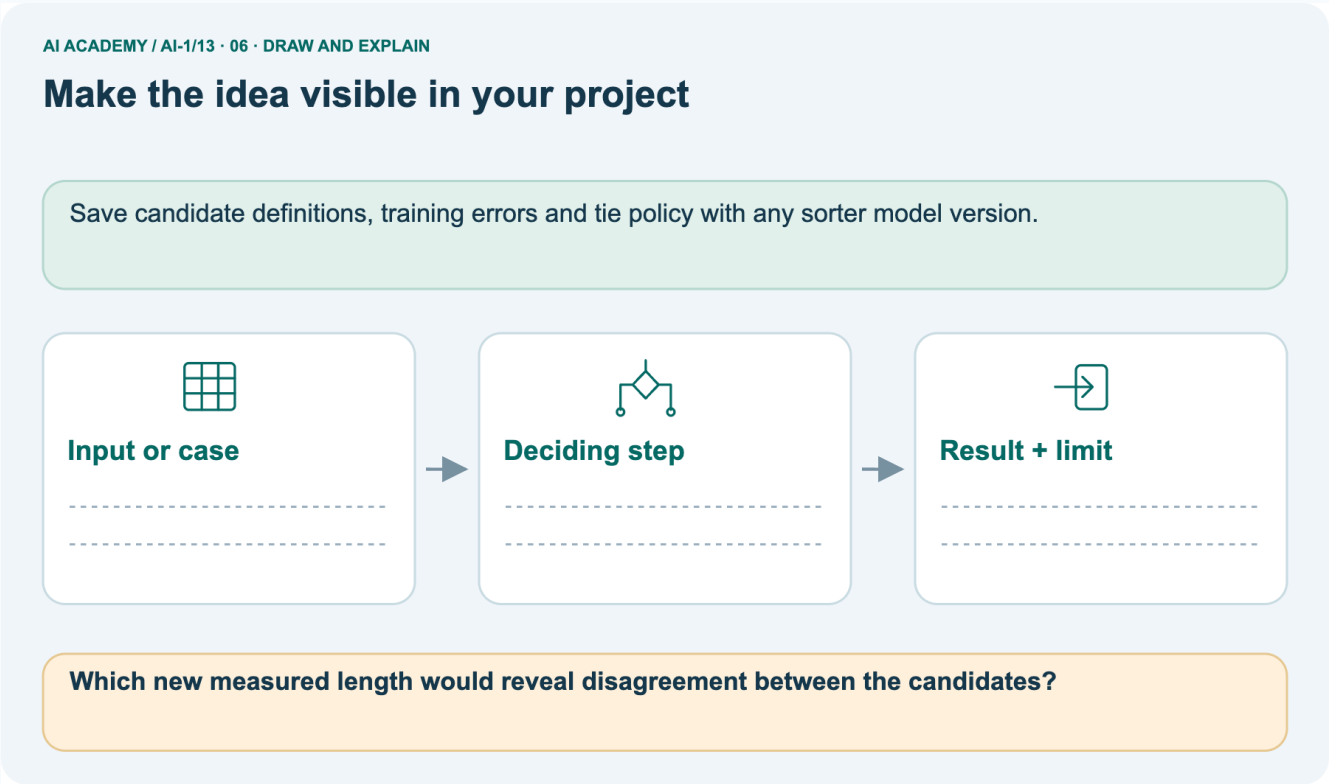
Follow the labelled picture with the matching explanation. Open the vector diagram

There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label. Limit: Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save candidate definitions, **training** errors and tie policy with any sorter model version. Draw your **input**, deciding step, result and limit.

Supported question: Which new measured length would reveal disagreement between the candidates?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Learning from Examples: the case

02. Follow the deciding steps

03. Same training errors, different new prediction

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.